

diffpair_test_routed

Design Report

kicad-tools 0.13.0

Rev 1 | 2026-05-13 | jlcpb

Board Summary

Property	Value
Layers	4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu)
Footprints	7 (0 SMD, 0 THT, 7 other)
Nets	26
Traces	130 segments
Vias	24
Board Size	100.0 x 80.0 mm

Design Overview

Theory of Operation

Differential Pair Test Board

Multi-protocol HSDI regression testbench

Epic #2556 Phase 4L (issue #2658)

Communication Interfaces

Protocol	Signals
UART	PCIE_TX+, USB3_RX1-
USB	USB2_D+, USB2_D-, VBUS_USB

Power Architecture

Power Rails: PWR_FLAG

Assembly Notes

- 4 fine-pitch components
- **Fine-pitch components:** 4 (U1, U2, U3, U4)

ERC Status

Metric	Count
Errors	0
Warnings	0

Status: SKIPPED – ERC skipped by user request

Bill of Materials

Value	Package	Qty	References
FFC4	FFC_4P_0.5mm	1	J4
MiniPCIe	PCIE_Mini_Edge	1	J3
USB-C	USB_C_Receptacle_USB2.0	1	J1
BGA49_USB3	BGA-49_5.0x5.0mm_Layout7x7_P0.5mm	1	U2
QFN24_MIPI	QFN-24-1EP_4x4mm_P0.5mm	1	U4
QFN32_USB2	QFN-32-1EP_5x5mm_P0.5mm	1	U1
QFP48_PCIe	LQFP-48_7x7mm_P0.5mm	1	U3

DRC Status

Metric	Count
Errors	32
Warnings	8
Blocking	32

Status: FAIL ### Violations by Type

Violation Type	Count
impedance	25
zone_unfilled	5
clearance_pad_segment	3
diffpair_clearance_intra	3
silkscreen_text_height	2
silkscreen_over_pad	1
via_in_pad	1

Manufacturing Readiness

Verdict: NOT_READY

Action Items

- **[CRITICAL]** Fix 32 blocking DRC violations (impedance (25), clearance__pad__segment (3), diffpair_clearance_intra (3))
- **[OPTIONAL]** Verify zone fill in KiCad: 2 nets appear incomplete but may be connected via zone fills
- **[OPTIONAL]** Verify zone fill in KiCad for 5 zone-connected nets
- **[OPTIONAL]** Review 8 DRC warnings

Routing Status

Metric	Value
Signal Net Completion	90.5% (19/21)
Overall Completion	84.6%
Complete Nets	22 / 26
Zone-Connected Nets	5
Incomplete Nets	4
Unconnected Pads	124

Zone-Connected Nets

- +1V2
- +1V8
- +3V3
- GND
- VBUS_USB

Unrouted Signal Nets

- USB3_TX1+
- USB3_TX1-

Unrouted Signal Nets

- USB3_TX1+
- USB3_TX1-

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~3.6 USD
Components (estimated)	~2.3 USD
Assembly (estimated)	~2.05 USD
Total (estimated)	~7.95 USD
Batch Quantity	5
Batch Total (estimated)	~39.74 USD